

What This Demo Shows

A one-command, reproducible governed run of the CONEXUS Sovereign pipeline using Gemini 2.0 Flash as the cloud LLM backend.

The pipeline's governance operators (Collapse, Become, Paradox-Hold, Observer) enforce structural invariants on LLM output after generation. This demo confirms those invariants hold identically whether the LLM is a local model or a cloud API.

The same invariants were previously verified on three local models (LLaMA 8B, Mistral 7B, Phi 4B). Gemini Flash is the fourth model to pass.

Headline Result

Invariant	Result
Open tensions after Collapse	0
Paradoxes held	94 / 94 (100%)
Paradoxes vetoed	94 / 94 (100%)
Observer attestations	3

Gemini is the fourth model to pass these governance invariants.

Four models. Different architectures. Different providers. Same structural outcome. The governance operators are model-agnostic.

Run Summary

Metric	Value
Model	gemini-2.0-flash (Google, cloud API)
Baseline	Sovereign-V5-Anchor (sealed)
Governance	v1
Operator sequence	Collapse > Become > Paradox-Hold > Observer
Duration	17.5 minutes
Gemini API calls	604
Final claims	862
Final tensions	1,602 (0 open)
Paradoxes	94 (100% held, 100% vetoed)

How To Run This Demo

Prerequisites: Python 3.10+, google-genai package, a Gemini API key (free tier sufficient).

Setup

```
export GEMINI_API_KEY="your-key"          # Linux/Mac
$env:GEMINI_API_KEY = "your-key"         # PowerShell
```

Run

```
python -m SovereignNEXT.pipeline.run_gemini_demo_v1 --passes 1 --seed 42
```

Expected runtime: 15-25 minutes. Governance outcomes (zero open tensions, 100% held, 100% vetoed) are deterministic for a given seed. Text content varies due to LLM non-determinism.

Verify

```
python verification/_verify.py
```

Expected output: VERIFIED - all hashes match.

What This Demo Does NOT Claim

This is not a product or hosted service.

This is not a performance benchmark.

This does not prove optimality or completeness.

This does not replace the Phase One three-model proof (sealed separately).

LLM output quality is not evaluated - only governance enforcement is tested.

Claim Boundary

This demo extends the governance invariance result to a fourth model (Gemini Flash, cloud API). It confirms the Sovereign pipeline's structural enforcement is independent of the LLM provider.

The LLM generates text. The governance operators enforce structure after generation. Different models produce different text, but the invariants are identical. The operators are model-agnostic. That is the proof.

Gate Report

Gate	Result
0 - Sealed paths verified before run	PASSED
1 - Adapter interface created	PASSED
2 - Gemini standalone test	PASSED
3 - Full governed cycle via Gemini	PASSED
3B - Governance invariants confirmed	PASSED
4A - OpenClaw Gateway bridge created	PASSED
4B - Route-via-OpenClaw flag	PASSED
5 - Demo proof packet sealed	PASSED
6 - 90-second demo script	PASSED
7 - Final report + sealed path verification	PASSED

All gates passed. No failures. No patches. All sealed Phase One paths hash-verified unchanged before and after execution.

Architecture (Single Pass)

1. Load sealed V5 baseline snapshot (650 claims, 1505 tensions, 84 paradoxes)
2. Become pass via Gemini: claim expansion + tension detection (604 API calls)
3. Collapse operator: commit or hold all tensions (0 left open)
4. Become operator: apply structural transforms
5. Paradox-Hold operator: enforce hold on all paradoxes (94/94 held, 94/94 vetoed)
6. Observer: attestation and anomaly detection (3 attestations)

Source Code

Repository: github.com/CONEXUS-dev/conexus-sovereign-ai

Demo runner: [SovereignNEXT/pipeline/run_gemini_demo_v1.py](#)

Gemini adapter: [SovereignNEXT/adapters/cloud_llm/gemini_client.py](#)

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